

MSc. in Computing Practicum Approval Form

Section 1: Student Details

Project Title:	Visual QA Chart
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Date of Submission	24-Nov-24

Section 2: About your Practicum

Visual Question Answering (VQA) is task that combines computer vision and natural language processing to answer questions about images. It enables more intuitive human-machine interaction by allowing users to ask questions about visual content. When it comes to interpreting structured visual data, such as charts and graphs, the field shifts toward ChartQA.

The objective of this project is to develop **Chart Question-Answering (ChartQA)** system capable of interpreting diverse chart types (e.g., bar charts, line charts, pie charts) and providing accurate context-aware answers to user queries with open vocabulary. This project leverages state-of-the-art machine learning models and publicly available datasets to train and evaluate the system.

Relevant Research Papers:

1. Ishmam, Md.F., Shovon, Md.S.H., Mridha, M.F., Dey, N., 2024. From image to language: A critical analysis of Visual Question Answering (VQA) approaches, challenges, and opportunities. Information Fusion 106, 102270. <https://doi.org/10.1016/j.inffus.2024.102270>
2. Kafle, K., Price, B., Cohen, S., Kanan, C., 2018. DVQA: Understanding Data Visualizations via Question Answering, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition. Presented at the 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, Salt Lake City, UT, pp. 5648–5656. <https://doi.org/10.1109/CVPR.2018.00592>
3. Kim, D.H., Hoque, E., Agrawala, M., 2020. Answering Questions about Charts and Generating Visual Explanations, in: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, CHI '20. Association for Computing Machinery, New York, NY, USA, pp. 1–13. <https://doi.org/10.1145/3313831.3376467>

4. Masry, A., Do, X.L., Tan, J.Q., Joty, S., Hoque, E., 2022. ChartQA: A Benchmark for Question Answering about Charts with Visual and Logical Reasoning, in: Muresan, S., Nakov, P., Villavicencio, A. (Eds.), Findings of the Association for Computational Linguistics: ACL 2022. Presented at the Findings 2022, Association for Computational Linguistics, Dublin, Ireland, pp. 2263–2279.
<https://doi.org/10.18653/v1/2022.findings-acl.177>
5. Methani, N., Ganguly, P., Khapra, M.M., Kumar, P., 2020. PlotQA: Reasoning over Scientific Plots. Presented at the Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 1527–1536.

How does your proposal relate to existing work on this topic described in these papers?

The research topic **Visual Chart QA** is fundamentally connected to the insights provided by the mentioned papers, as they collectively address the challenges and advancements in enabling machines to understand and reason over visual data. The (Methani et al., 2020) and (Masry et al., 2022) papers are especially relevant, as they focus on creating datasets with real-world charts and complex reasoning questions, going beyond synthetic and template-based questions. These datasets emphasize the need for models that combine visual features, numeric reasoning, and open-vocabulary processing, which is central to advancing VQA for graphs and charts. Similarly, the (Kafle et al., 2018) paper highlights the importance of handling dynamic and unique chart-specific vocabularies, a critical requirement for real-world applications such as scientific reports and business dashboards.

The paper (Kim et al., 2020) adds another layer of relevance by exploring explainability in VQA systems. It demonstrates how providing visual explanations alongside answers can enhance user trust and comprehension, a crucial aspect for deploying VQA systems in practical scenarios. Additionally, the broader analysis in (Ishmam et al., 2024) situates chart-based VQA within the evolving field of multimodal learning, showcasing the shift to vision-language pretraining and transformer-based models. Collectively, these papers provide the datasets, methods, and challenges that directly inform and shape the research agenda in VQA for charts and graphs.

What are the research questions that you will attempt to answer?

How to enhance reasoning and handle open-vocabulary answers?

- Enhance reasoning and handle open-vocabulary answers by leveraging transformer-based models and fine-tuning on diverse datasets with complex queries.

How to integrate visual and tabular data for accurate QA?

- Integrate visual and tabular data by combining multi-modal architectures with attention mechanisms to align visual elements with textual/tabular information.

How to ensure explainability across diverse chart types?

- Ensure explainability by incorporating interpretable model outputs and training on diverse and varied chart types.

- What software and programming environment will you use?

We will use Python as the primary programming language. Development will be conducted on Jupyter Notebook and Google Colab.

- What coding/development will you do?

We will develop code for feature extraction from charts, fine-tune pre-trained models for question answering, and build an end-to-end pipeline for processing and responding to user queries.

- What data will be used for your investigations?

Below mentioned dataset cover wide range of chart types and question styles.

PlotQA- A large-scale dataset containing synthetic charts and natural language questions.

ChartQA- A dataset designed specifically for Chart Question Answering, containing real-world charts and diverse question types.

FigureQA - A dataset of over 180,000 figures (charts and graphs) with QA pairs focused on visual reasoning.

- Is this data currently available, if not, where will it come from?

Yes, datasets are publicly available. If additional data is required, it will be collected through web scraping or using visualization libraries like Matplotlib for synthetic charts.

- What experiments do you expect to run?

We expect to run experiments on feature extraction by testing OCR(**Optical Character Recognition**) systems for accurate label and axis information retrieval from charts. Model learning experiments will involve training models that integrate visual and textual data.

- What output do you expect to gather?

We expect to get correct answer for open-vocabulary and complex reasoning questions.

- How will the results be evaluated?

To evaluate open-vocabulary results in ChartQA, we will use a combination of automated metrics and human evaluation. Automated metrics assess overlap with reference answers, while human evaluation judges correctness, relevance, clarity, and completeness.